INCH-POUND

MIL-H-48358 (AR)
AMENDMENT 3
26 August 1992
SUPERSEDING
AMENDMENT 2
8 April 1988

MILITARY SPECIFICATION

HMX/RESIN EXPLOSIVE COMPOSITION
LX-14-0
(For Use in Ammunition)

This Amendment forms a part of Military Specification MIL-H-48358 (AR), dated 30 September 1977, and is approved for use by the US Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

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Table I - after MIL-P-63196 add "Type I"

* Para 3.1.1 Table I - after MIL-P-63196 add "Type I or III".

PAGE 3

3.2.3 Granulation. Correct "compsition" to "composition": and "specified-in" to "specified in".

PAGE 4

3.4 Foreign matter. Correct "woods" to "wood".

4.3.1 Submission. Add "The sample shall be obtained from a production lot which has been produced by the contractor using the same production processes, procedures, and equipment as will be used in fulfilling the contract. All materials shall be obtained from the same sources of supply as will be used during regular production."

4.3.3 Rejection. Add "See MIL-A-48078".

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* Para 4.4.1.b - Delete in its entirety and substitute:

"4.4.1.b - Binder from one manufacturer only. Not more than two blended acceptable binder lots of the same type produced in accordance with the same specification or specification revision under one continuous set of operating conditions. Mixing of Types I and III are not permitted."

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4.5 Testing. Add "PRECAUTION - This specification covers sampling and testing of toxic and hazardous materials. Accordingly, it is emphasized that all applicable safety rules, regulations, and procedures must be followed in the handling and processing of these materials."

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4.6.3.1 HMX Content. Delete 4.6.3.1.1 and 4.6.3.1.2 in their entirety and substitute the following:

"4.6.3.1.1 Reagent. Prepare the extraction solvent by adding excess RDX and HMX to 1, 2-dichloroethane (EDC). The mixture shall be maintained for at least eight hours at the temperature (± 2 degrees C) at which the analysis will be performed. Prior to use for extraction, the EDC solution shall be agitated for two hours and filtered.

4.6.3.1.2 Procedure. Approximately 2.5g of the dried sample shall be accurately weighed (recorded to the nearest 0.1 mg) and quantitatively transferred to a 250 ml screw cap bottle. Add 150 ml of saturated EDC solution (see 4.6.3.1.1), attach the screw cap, and place the bottle on a mechanical shaker. The bottle shall be shaken at least thirty minutes, until visual examination indicates complete dissolution of the elastomeric binder. The sample solution shall then be transferred to a tared filtering crucible attached to a vacuum flask. About 30 ml of the saturated EDC solution shall then be poured into the bottle, the screw cap replaced, the bottle shaken by hand, and the contents transferred to the filtering crucible. This procedure shall be repeated at least twice in order to remove all residue from the bottle. The bottle and cap shall then be rinsed twice with 30 ml aliquots of the EDC solution, with the washings being transferred to the filtering crucible. The crucible shall then be removed from the flask, 20 ml of the saturated EDC solution shall be added, and the crucible shall be reconnected to the flask, allowing the liquid to filter through the residue. This procedure shall be performed a total of three times. The crucible shall then be removed from the flask and the interior portions beneath the glass frit shall be carefully rinsed with the saturated EDC. After rinsing, the crucible shall again be connected to the vacuum flask and aspirated for approximately two minutes. The crucible shall then be placed in an oven (100 ± 5 degrees C) for one hour, or until dry. After cooling to room temperature the crucible shall be accurately weighed (recorded to the nearest 0.1 mg). The HMX content of the sample shall be calculated as shown below:

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$$\% \text{ HMX} = \frac{100\ (A-B)}{W}$$

A = Final weight of crucible, g

B = Tare weight of crucible, g

W = Sample weight, g"

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Delete section 5 in its entirety and replace with:

"5. PREPARATION FOR DELIVERY

5.1 PACKING

5.1.1 LEVEL A Packing shall be in accordance with Dwg. 7548644.

5.1.2 LEVEL B Packing shall be in accordance with Level A packing requirements.

5.1.3 LEVEL C Packing shall be in accordance with Dwg. 7548645 or Dwg. 9257923.

5.2 MARKING

5.2.1 LEVEL A Mark in accordance with Dwg. 7548644.

5.2.2 LEVEL B Mark in accordance with Level A marking requirements

5.2.3 LEVEL C Mark in accordance with Dwg. 7548645 or Dwg. 9257923."

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6.2: Delete all material following "DRDAR-QA, Dover, New Jersey".

6.2: Delete "DRDAR-QA, Dover, New Jersey" and substitute "AMSMC-QAR-R(D), Picatinny Arsenal, NJ 07806-5000".

6.3: Change "5703 F1" to "5702 F1 (MIL-P-63196 Type I)" and delete "in accordance with MIL-P-63196".

* 6.3: Change "5702F1 (MIL-P-63196 Type I)" to: "... 5702 (MIL-P-63196 Type I) or 5703P (MIL-P-63196 Type III)...".

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* 6.6: Change the first sentence in the paragraph from "Heliogen Violet Toner ... found satisfactory" to "Heliogen Violet Toner 49-6001 manufactured by General Aniline and Film Corp. and Paliogen Violet Toner VT 4414 manufactured by BASF Wyandotte have been found satisfactory."

The margins of this amendment are marked with an asterisk or vertical lines to indicate where the changes (additions, modification, corrections, deletions) from the previous amendment were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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